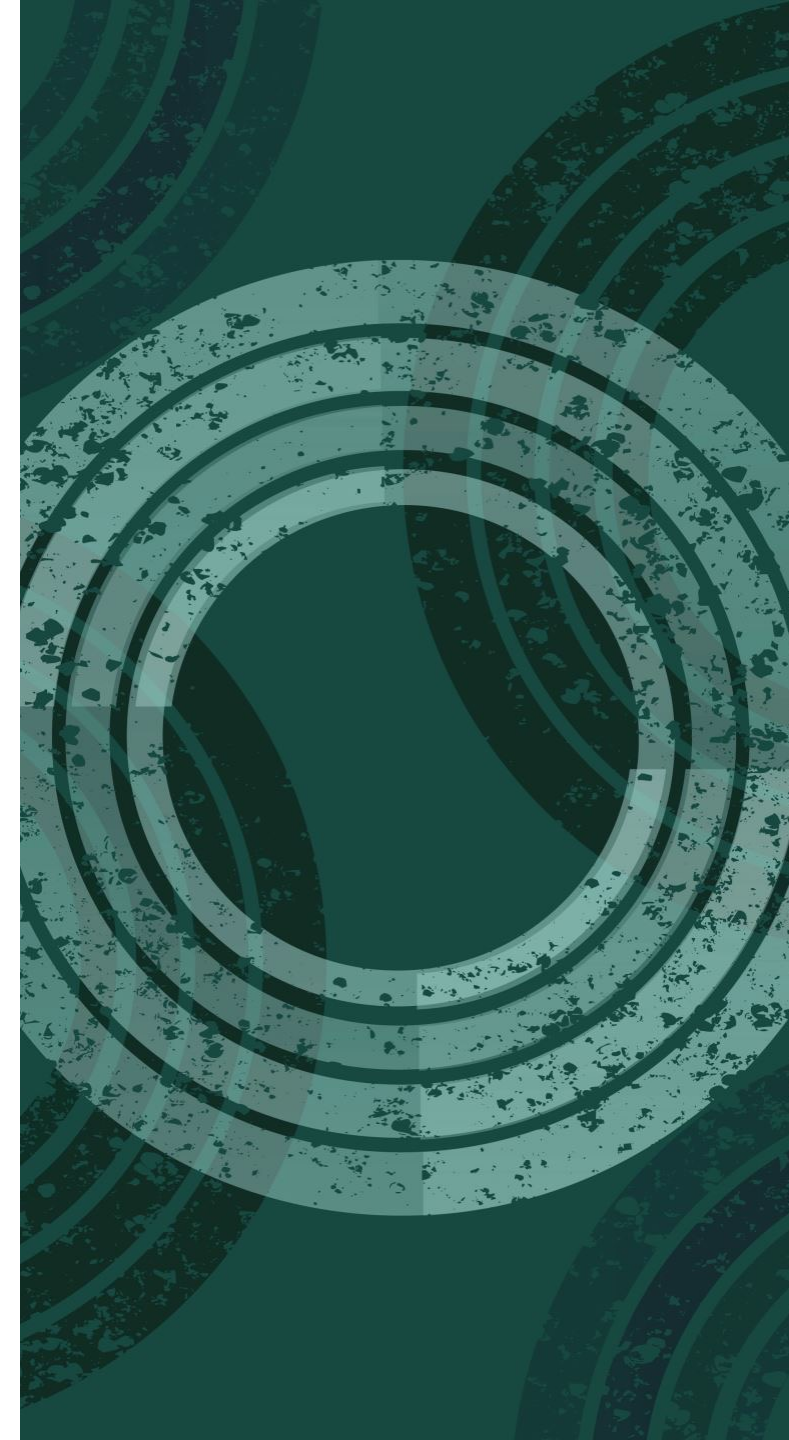
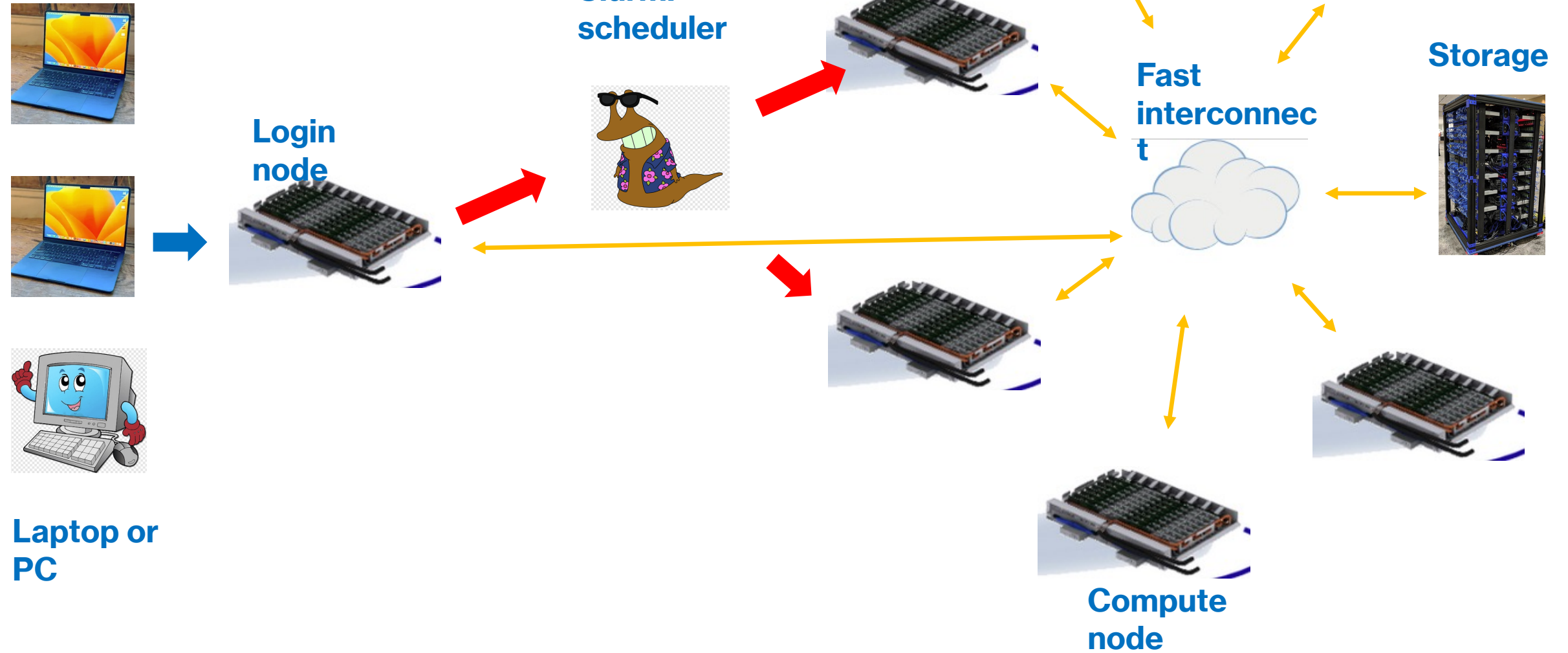


— INTRODUCTION TO GPU PARTITIONS ON ALPINE

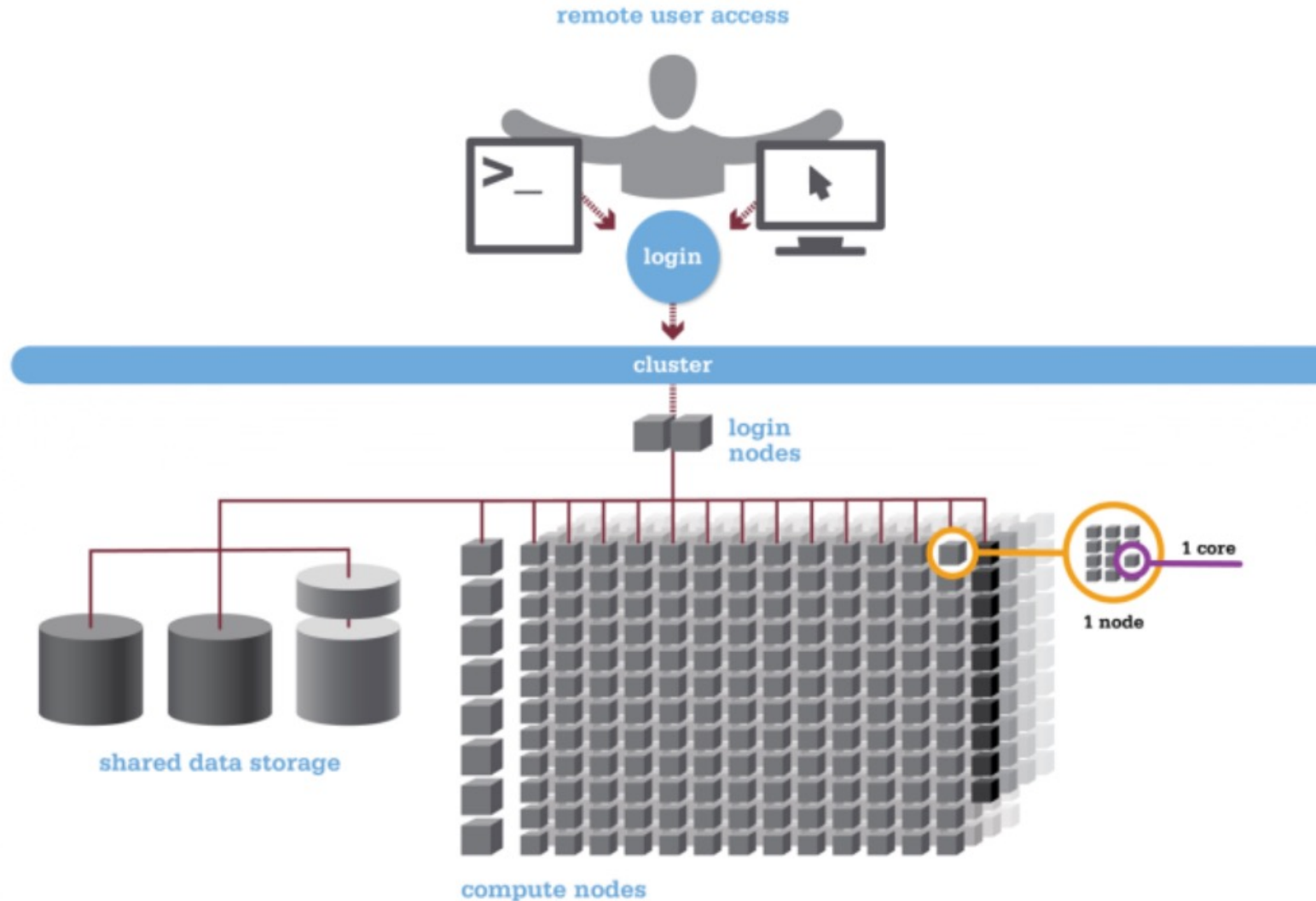
By Kevin Fotso



Anatomy of supercomputing



Architecture of a supercomputer

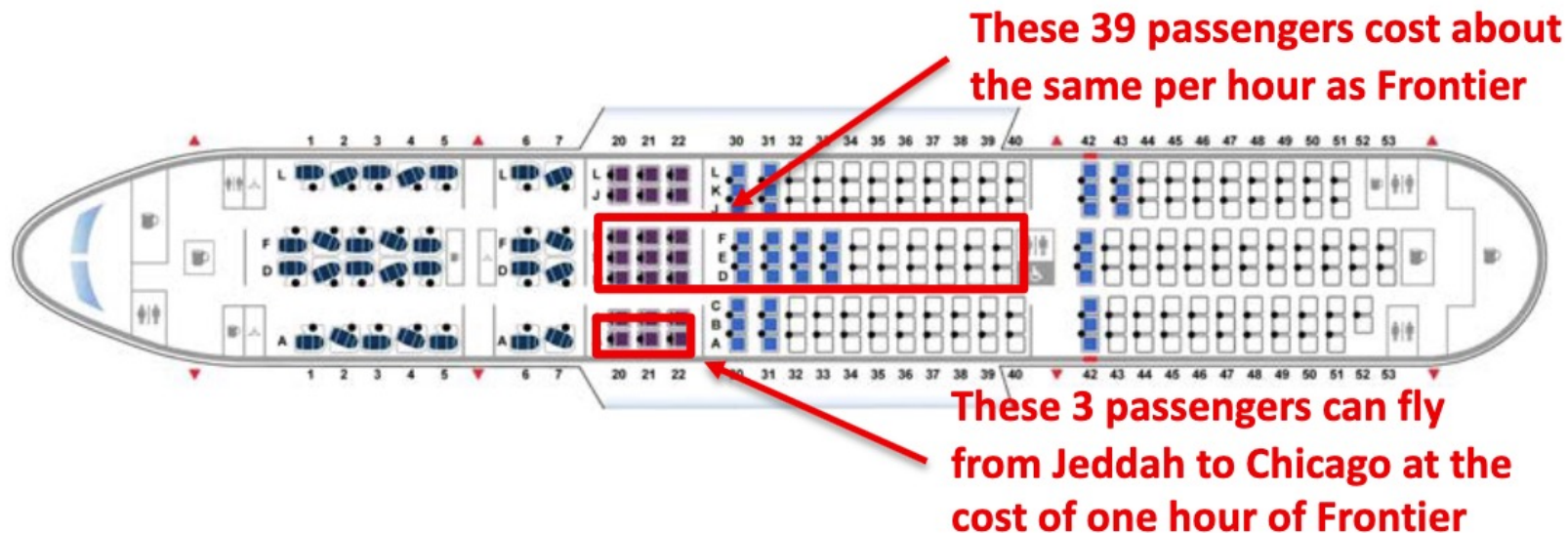


- Login nodes. To log into the system, cd into directories, look at files etc ...
- Compute node. Dedicated to do the computation
- The slurm scheduler controls access to the compute nodes to avoid a tragedy of the commons ...

Why Slurm?

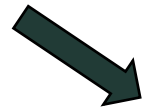


- Allows for better tracking of resource used per user
- Better management of of resource across user base
- More sustainable and less tragedy of the commons
- Widely used and supported by applications in case of bugs.



Why Slurm?

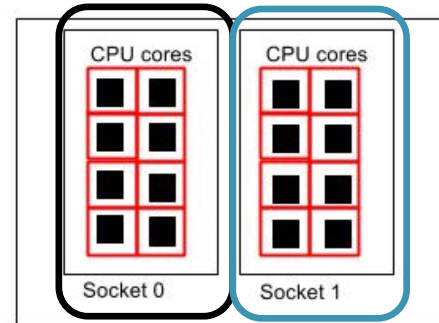
- Bulma request 8 cores from Node 1
- Dr. Brief requests 8 cores as well from Node 1
- Gohan requests 16 cores from Node 2
- No tragedy of the common!!!



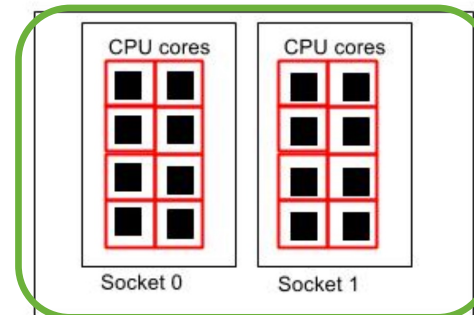
Slurm
scheduler



Node 1



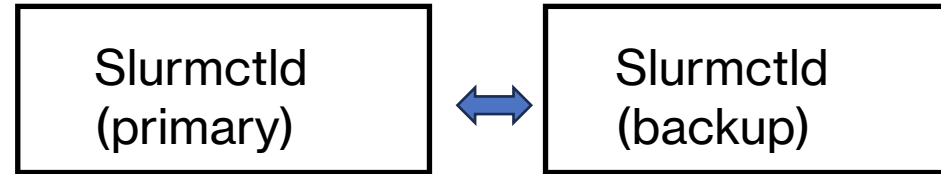
Node 2



Slurm architecture

- Slurmctld are daemons that monitor the state of resources, decide when to initiate job states and process users command.

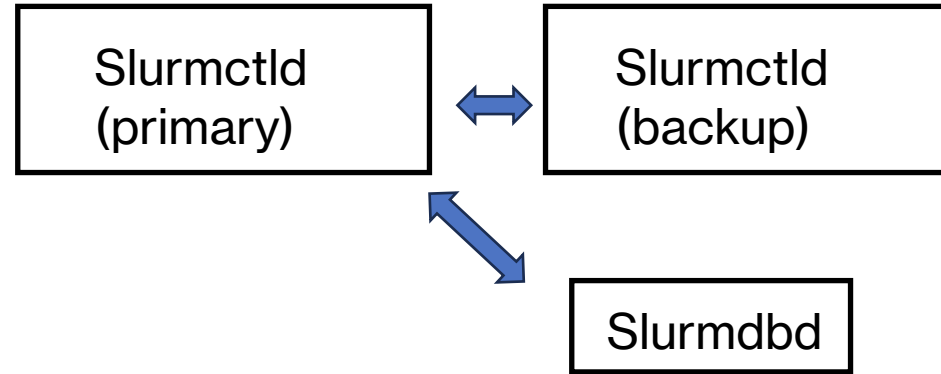
Controler daemons



Slurm architecture

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- Slurmdbd is the database daemon and records accounting information and manages some other config (qos, fair share ...)

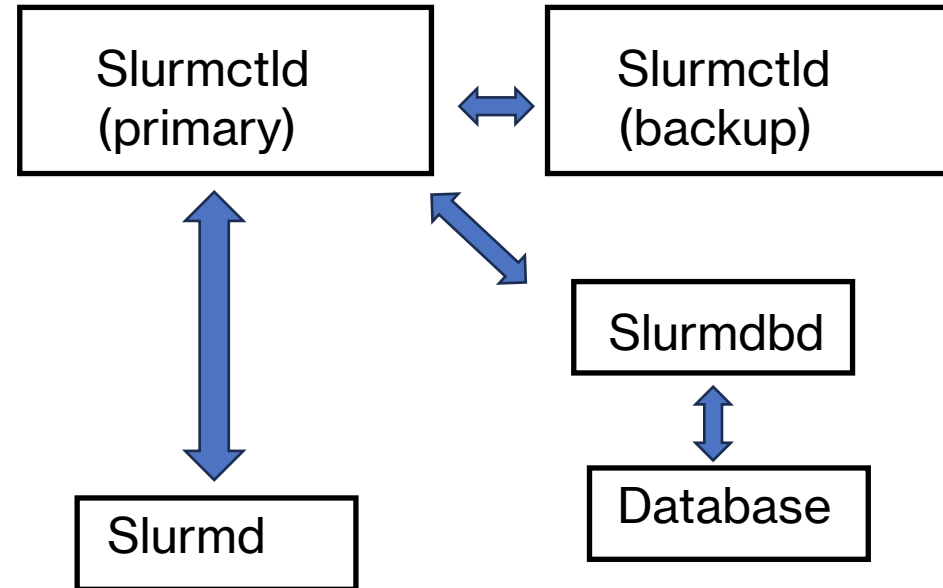
Controler daemons



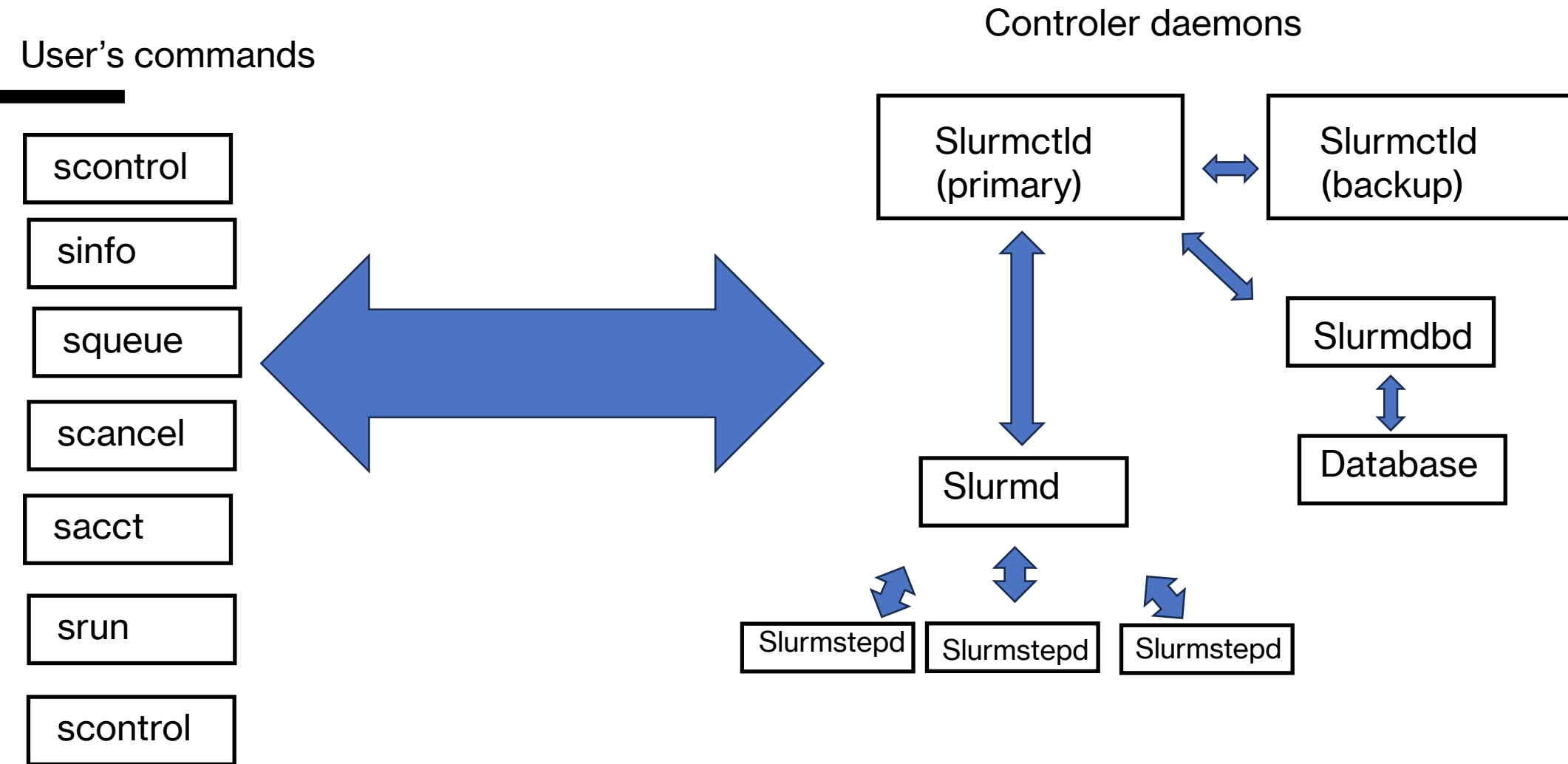
Slurm architecture

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- Slurmdbd is the database daemon and records accounting information and manages some other config (qos, fair share ...).
- Slurmd is the slurm compute daemon on compute nodes. Spawn processes on behalf of users

Controler daemons



Slurm architecture



Context[1]

- Our cluster named Alpine is made of 28,144 cores and 456 compute nodes

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- 3 gpus per node, except for Grace Hopper (1 gpu per node).
- The OS is RedHat 8.10.
- One needs to submit a proposal to access Grace Hopper. Let me know if interested!

Software stack

- Our software versions on the cuda side are: [cuda/11.2](#), [cuda/11.3](#), [cuda/11.4](#), [cuda/11.8](#) and [cuda/12.1.1](#).

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- Our software versions on the AMD side are: [rocm/5.2.3](#), [rocm/5.3.0](#), [rocm/5.5.0](#), [rocm/5.6.0](#), [rocm/6.1.0](#)

QOS changes on Alpine

- To improve resource availability across Alpine GPU partitions and simplify job submissions, several QOS changes have been made.

QOS changes on Alpine

Quality of service (QOS) type	Description
normal	For a job with a wall time of up to 24 hours and in batch mode.
long	A job with a walltime between 24 hours and 7 days and in batch mode.
testing	For jobs that interact with a gpu interactively and not in batch mode



QOS types before the July maintenance.

QOS changes on Alpine

- The new QOS combines the old qos definition with the generic resource (GRES) type in order to reduce very long wait times.

QOS changes on Alpine

Quality of service (QOS) type	Description
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QOS types **after** the July maintenance.

Recap

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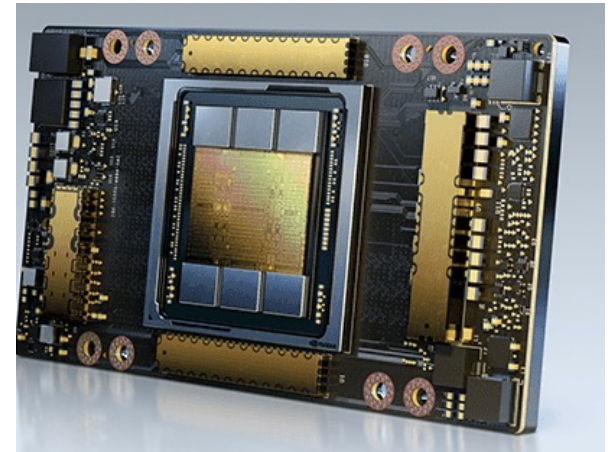
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Computing situation

- Our users on the Medical Campus side have been used to computation on the A100 nodes, and more recently L40 nodes.

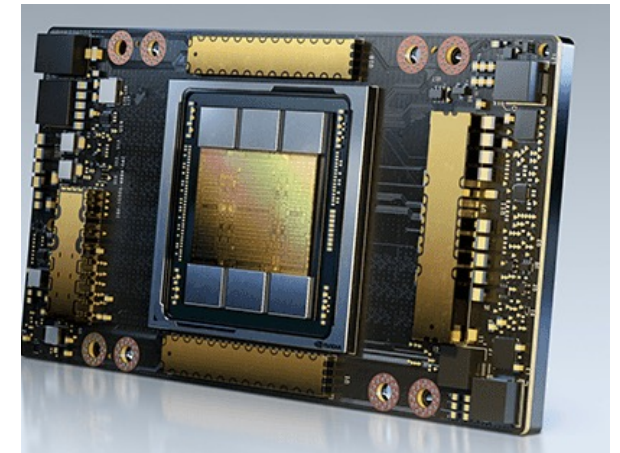
Alpine NVIDIA A100 Specs

- 40GB or 80 GB global memory for each gpu.



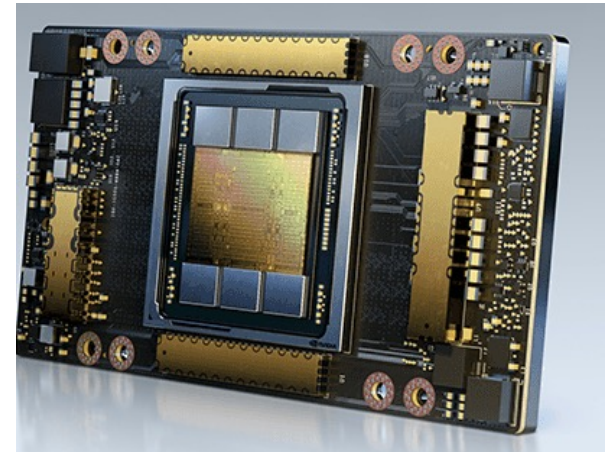
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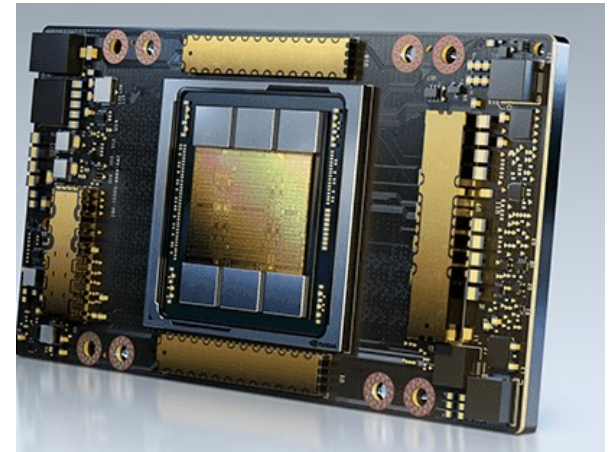
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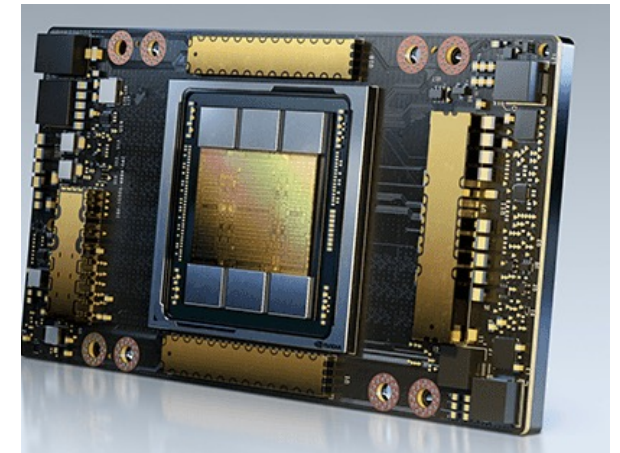
- 40GB or 80 GB global memory for each gpu.
- Peak memory bandwidth is >2TB/s.
- 3 NVIDIA gpus per node
- One partition name on Alpine: aa100 for both batch submission and interactive debugging.



Alpine NVIDIA A100 Specs

To access the A100 compute node with 40G please include the correct partition and gres types as follows:

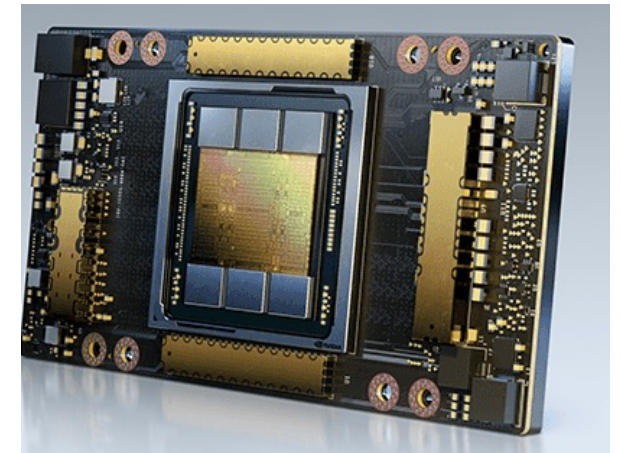
- **#SBATCH --partition=aa100**
#SBATCH --gres=gpu:a100-40gb:1
- **#SBATCH --qos=gpu-normal**



Alpine NVIDIA A100 Specs

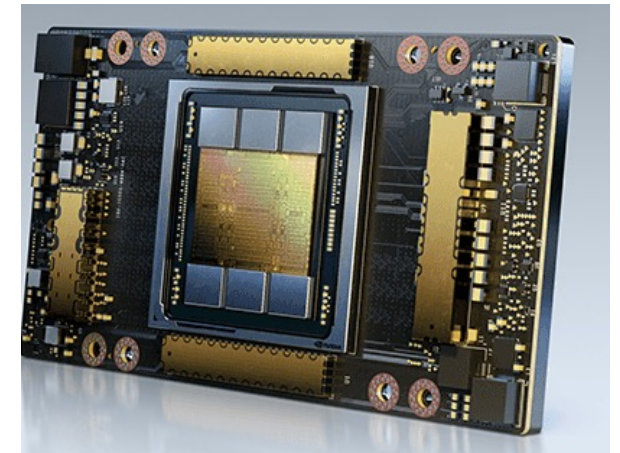
Similarly, to access the A100 compute node with 80G please include the correct partition and gres types as follows:

- **#SBATCH --partition=aa100**
#SBATCH --gres=gpu:a100_80gb:1
- **#SBATCH --qos=gpu-normal**



Alpine NVIDIA L40 Specs

- 48GB global memory for each gpu.
- Peak memory bandwidth is 864 GB/s.
- 3 NVIDIA gpus per node
- One partition names on Alpine: al40 for batch submission



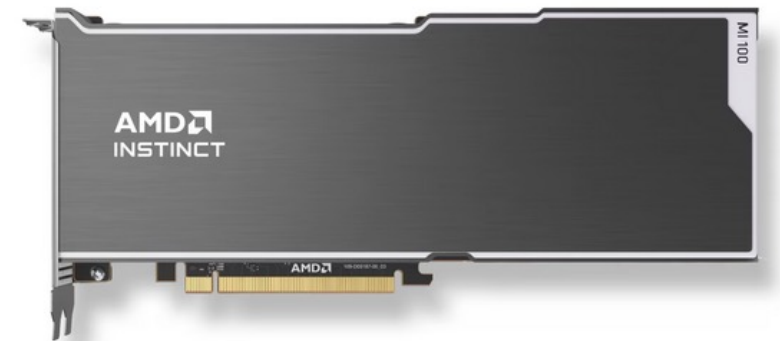
Alpine NVIDIA L40 Specs

To access the L40 compute node with 48G please include the correct partition and gres types as follows:

- **#SBATCH --partition=al40**
#SBATCH --gres=gpu:l40:1
#SBATCH --qos=gpu-normal

Alpine AMD MI100 Specs

- 32 GB global memory for each gpu (up to 80 GB memory for the NVIDIA A100 gpu)



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- 3 AMD gpus per node



Alpine AMD MI100 Specs

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- 3 AMD gpus per node
- One partition names on Alpine: ami100 for batch submission.



Alpine AMD MI100 Specs

- To access the ami100 compute node with 32G please include the correct partition and gres types as follows:
 - **#SBATCH --partition=ami100**
 - #SBATCH --gres=gpu:mi100:1**
 - **#SBATCH --qos=gpu-normal**



Slurm & GPU partitions

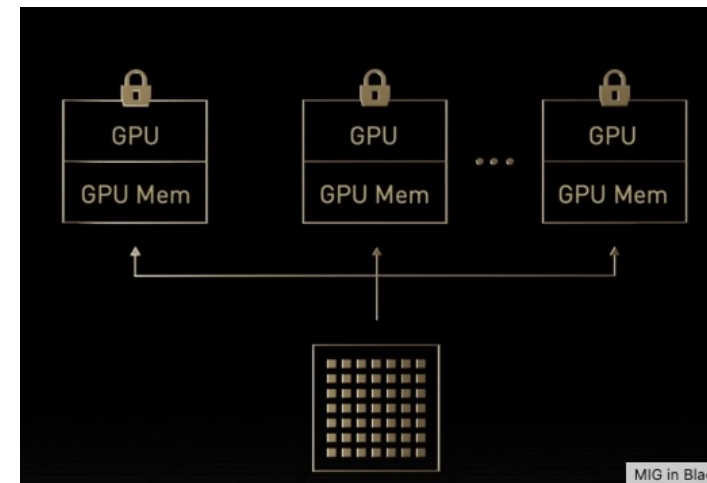
- `sinfo --Format partition`
- `sinfo --partition aa100,ami100 --Format=Partition,Nodes,Time`
- `scontrol show partition aa100`
- `scontrol show node c3gpu-c2-u13`



To print current state, type of nodes, default walltime, partition name and information, total nodes etc ...

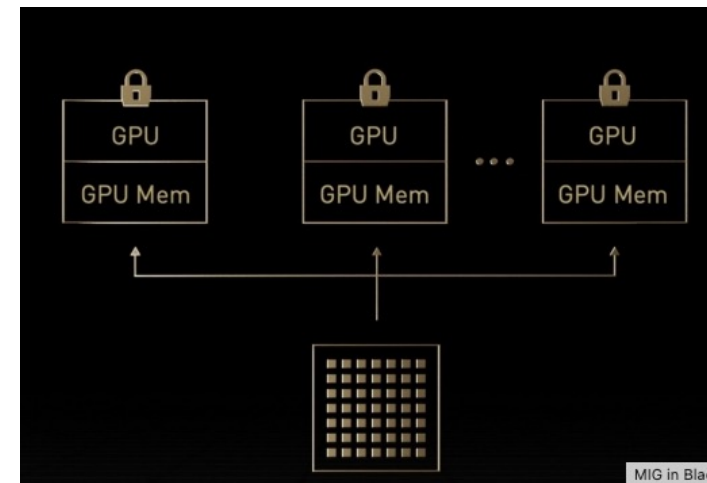
NVIDIA MIG

- The aa100 partition utilizes NVIDIA's [Multi-Instance GPU \(MIG\)](https://docs.nvidia.com/datacenter/tesla/mig-user-guide/latest/concepts.html) feature, which can “slice” GPUs into multiple GPU instances: <https://docs.nvidia.com/datacenter/tesla/mig-user-guide/latest/concepts.html>



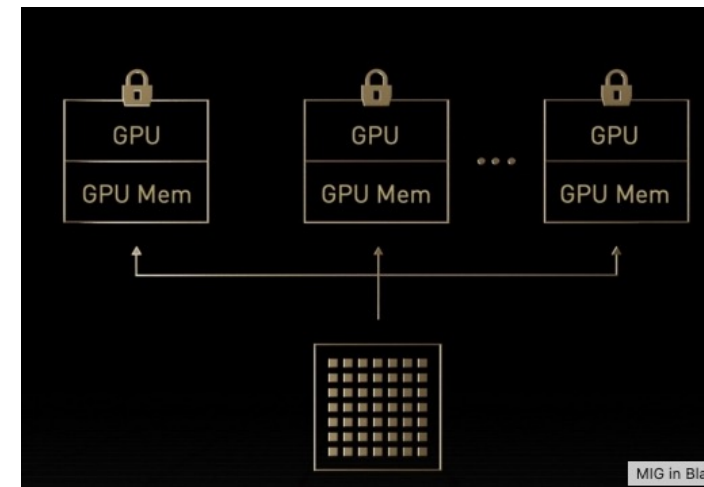
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NVIDIA MIG

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- These GPU instances can be treated as a single GPU.
- One important limitation is that MIG does not allow for multiple GPU instances to communicate with each other.



NVIDIA MIG

- To access a gpu for debugging purposes, most gres call will adopt a structure similar to this:
- `--gres=gpu:<gpu_name>_compute_slice.memory_slice:total_gpus_requested`

NVIDIA MIG

- For example:

`--gres=gpu:a100_3g.20gb:1.`



Generic
resources

NVIDIA MIG

- For example:

--gres=gpu:100_3g.20gb:1.



Compute
slice

NVIDIA MIG

- For example:

`--gres=gpu:a100_3g.20gb:1.`



Memory
slice

NVIDIA MIG

- Below are the full documented changes to access a gpu for debugging (interactive) purposes:

Before the July maintenance	After the July maintenance
<code>sinteractive --partition=atesting_a100 --gres=gpu:1 --qos=testing --time=00:30:00 --nodes=1 --ntasks=4</code>	<code>sinteractive --partition=aa100 --gres=gpu:a100_3g.20gb:1 --qos=gpu-normal --time=00:30:00 --nodes=1 --ntasks=4</code>
<code>sinteractive --partition=atesting_ami100 --gres=gpu:1 --qos=testing --time=00:30:00 --nodes=1 --ntasks=4</code>	<code>sinteractive --partition=ami100 --gres=gpu:mi100:1 --qos=gpu-normal --time=00:30:00 --nodes=1 --ntasks=4</code>

Full slurm scripts

Documented changes on how to have access to an a100 (40G) gpu in slurm batch mode.

Before:

```
#!/bin/bash
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --nodes=1
#SBATCH --account=amc-general
#SBATCH --partition=aa100
#SBATCH --time=00:20:00
#SBATCH --output=FFT.%j.out
#SBATCH --error=FFT.%j.err
#SBATCH --qos=normal
#SBATCH --gres=gpu:1
```



Now:

```
#!/bin/bash
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --nodes=1
#SBATCH --account=amc-general
#SBATCH --partition=aa100
#SBATCH --time=00:20:00
#SBATCH --output=FFT.%j.out
#SBATCH --error=FFT.%j.err
#SBATCH --qos=gpu-normal
#SBATCH --gres=gpu:a100-40gb
```

Full slurm scripts

Documented changes on how to have access to an a100 (80G) gpu in slurm batch mode.

Before:

```
#!/bin/bash
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --nodes=1
#SBATCH --account=amc-general
#SBATCH --partition=aa100
#SBATCH --time=00:20:00
#SBATCH --output=FFT.%j.out
#SBATCH --error=FFT.%j.err
#SBATCH --qos=normal
#SBATCH --gres=gpu:1
```



Now:

```
#!/bin/bash
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --nodes=1
#SBATCH --account=amc-general
#SBATCH --partition=aa100
#SBATCH --time=00:20:00
#SBATCH --output=FFT.%j.out
#SBATCH --error=FFT.%j.err
#SBATCH --qos=gpu-normal
#SBATCH --gres=gpu:a100_80gb
```

Full slurm scripts

Documented changes on how to have access to an L40 (48G) gpu in slurm batch mode.

Before:

```
#!/bin/bash
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --nodes=1
#SBATCH --account=amc-general
#SBATCH --partition=al40
#SBATCH --time=00:20:00
#SBATCH --output=FFT.%j.out
#SBATCH --error=FFT.%j.err
#SBATCH --qos=normal
#SBATCH --gres=gpu:1
```



Now:

```
#!/bin/bash
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --nodes=1
#SBATCH --account=amc-general
#SBATCH --partition=al40
#SBATCH --time=00:20:00
#SBATCH --output=FFT.%j.out
#SBATCH --error=FFT.%j.err
#SBATCH --qos=gpu-normal
#SBATCH --gres=gpu:l40:1
```

Full slurm scripts

Documented changes on how to have access to an MI100 (32G) gpu in slurm batch mode.

Before:

```
#!/bin/bash
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --nodes=1
#SBATCH --account=amc-general
#SBATCH --partition=ami100
#SBATCH --time=00:20:00
#SBATCH --output=FFT.%j.out
#SBATCH --error=FFT.%j.err
#SBATCH --qos=normal
#SBATCH --gres=gpu:1
```



Now:

```
#!/bin/bash
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --nodes=1
#SBATCH --account=amc-general
#SBATCH --partition=ami100
#SBATCH --time=00:20:00
#SBATCH --output=FFT.%j.out
#SBATCH --error=FFT.%j.err
#SBATCH --qos=gpu-normal
#SBATCH --gres=gpu:mi100:1
```

Best practices

What happens if my gpu deep learning job runs out of memory?

- Assess your network size based on the number of parameters and the batch size, then multiply it by the precision type.

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- For example, if your network is using **271,210** parameters with **float64** precision and **1** batch size, the calculation goes as follows:

[Mem_size= **271,210** parameters **X 8** bytes **X 1** batch size]

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- Reduce batch size.
- Consider doing distributed computing.
- Consider doing feature selection prior to running your model in order to reduce the size of your data.

What happens if my gpu deep learning job runs out of memory?

- If you are using data loaders with Pytorch, consider increasing the number of workers:

`(#SBATCH --cpus-per-task=10 #SBATCH --ntasks=1).`

- If you are using Pytorch, consider also adjusting the garbage collection and expandable segments parameters as follows:

```
os.environ["PYTORCH_CUDA_ALLOC_CONF"] = (  
    "expandable_segments:True,"  
    "garbage_collection_threshold:0.9"  
)
```

Monitoring tools (NVIDIA gpus)

- Nvtop (pip install nvtop)
- gpustats (pip install gpustats)
- nvidia-smi

NVITOP 1.4.2				Driver Version: 570.124.06		CUDA Driver Version: 12.8	
GPU Fan	Name Temp Perf	Persistence-M Pwr:Usage/Cap	Bus-Id	Disp.A Memory-Usage	MIG M. GPU-Util	Uncorr. ECC Compute M.	
0 N/A	A100-PCIE-40GB 29C P0	On 33W / 250W	00000000:21:00.0	Off 212.9MiB / 40.00GiB	Enabled N/A	0 Default	MEM: 0.5% UTL: N/A
0:0	3g.20gb @ GI/CI: 2/0		106.4MiB / 20096MiB		BAR1:	64KiB / 0%	MEM: 0.5%
1 N/A	A100-PCIE-40GB 28C P0	On 34W / 250W	00000000:81:00.0	Off 212.9MiB / 40.00GiB	Enabled N/A	0 Default	MEM: 0.5% UTL: N/A
2 N/A	A100-PCIE-40GB 29C P0	On 35W / 250W	00000000:E2:00.0	Off 212.9MiB / 40.00GiB	Enabled N/A	0 Default	MEM: 0.5% UTL: N/A
Load Average: 0.88 0.23 0.08 CPU: 0.4%							AVG GPU MEM: 0.5%
</							

kfotso@xsede.org@c3gpu-c2-u13.rc.int.colorado.edu

Run slurm script (NVIDIA GPU)

- sbatch **horovod_mpi.sh**
- Check the nodes where the job is running

```
~$ squeue -u kfotso@x
JOBID PARTITION NAME USER ST TIME NODES NODELIST(REASON)
81362 aa100 horovod_ kfotso@x R 2:34 2 c3gpu-a9-u31-1,c3gpu-a9-u31-2
79946 acompile acompile kfotso@x R 2:03:48 1 c3cpu-c9-u5-2
81778 atesting_ sinterac kfotso@x R 7:56 1 c3gpu-a9-u29-1
```

```
~$ squeue --me
JOBID PARTITION NAME USER ST TIME NODES NODELIST(REASON)
81362 aa100 horovod_ kfotso@x R 2:34 2 c3gpu-a9-u31-1,c3gpu-a9-u31-2
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81778 atesting_ sinterac kfotso@x R 7:56 1 c3gpu-a9-u29-1
```

```
~$ ssh c3gpu-a9-u29-1^C
```



We have a GPU
node that we can
ssh to:
c3gpu-a9-u29-1

GPU monitoring (NVIDIA GPU)

- ssh to a that node on a different screen and run:

`nvidia-smi -l`

NVIDIA-SMI 525.85.12 Driver Version: 525.85.12 CUDA Version: 12.0									
GPU	Name	Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr.	ECC		
Fan	Temp	Perf	Pwr:Usage/Cap	Memory-Usage	GPU-Util	Compute	M.		
							MIG		
0	NVIDIA A100	80G...	On	00000000:25:00.0	Off	0			
N/A	33C	P0	80W / 300W	1869MiB / 81920MiB	26%	Disabled			
1	NVIDIA A100	80G...	On	00000000:81:00.0	Off	0			
N/A	32C	P0	81W / 300W	1869MiB / 81920MiB	25%	Default			
						Disabled			

Processes:								
GPU	GI	CI	PID	Type	Process name	GPU Memory		
	ID	ID				Usage		
0	N/A	N/A	569301	C	python	1866MiB		
1	N/A	N/A	569302	C	python	1866MiB		

We can confirm
that we are using
2 GPUs!!!

GPU monitoring (NVIDIA GPU)

- ssh to a that node on a different screen and run:

```
nvidia-smi -l
```

NVIDIA-SMI 525.85.12 Driver Version: 525.85.12 CUDA Version: 12.0									
GPU	Name	Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr.	ECC		
Fan	Temp	Perf	Pwr:Usage/Cap	Memory-Usage	GPU-Util	Compute	M.		
						MIG	M.		
0	NVIDIA A100	80G...	On	00000000:25:00.0	Off	0			
N/A	33C	P0	80W / 300W	1869MiB / 81920MiB	26%	Disabled			
1	NVIDIA A100	80G...	On	00000000:81:00.0	Off	0			
N/A	32C	P0	81W / 300W	1869MiB / 81920MiB	25%	Default			
						Disabled			

Percentage GPU usability. That is the portion of time 1 or more kernels were used

Processes:								
GPU	GI	CI	PID	Type	Process name	GPU Memory Usage		
ID	ID	ID						
0	N/A	N/A	569301	C	python	1866MiB		
1	N/A	N/A	569302	C	python	1866MiB		

GPU monitoring (NVIDIA GPU)

- nvidia-smi has more parameters that can be used. Cheatsheet here:

<https://www.docs.arc.vt.edu/usage/gpumon.html>

```
[xsede.org@c3gpu-a9-u29-1 ~]$ nvidia-smi --query-gpu=timestamp,name,pci.bus_id,driver_version,temperature.gpu,utilization.gpu,utilization.memory --format=csv -l 5
timestamp, name, pci.bus_id, driver_version, temperature.gpu, utilization.gpu [%], utilization.memory [%]
2023/10/23 21:42:45.228, NVIDIA A100 80GB PCIe, 00000000:25:00.0, 525.85.12, 32, 0 %, 0 %
2023/10/23 21:42:45.230, NVIDIA A100 80GB PCIe, 00000000:81:00.0, 525.85.12, 31, 0 %, 0 %
2023/10/23 21:42:50.233, NVIDIA A100 80GB PCIe, 00000000:25:00.0, 525.85.12, 33, 7 %, 0 %
2023/10/23 21:42:50.234, NVIDIA A100 80GB PCIe, 00000000:81:00.0, 525.85.12, 32, 5 %, 0 %
2023/10/23 21:42:55.236, NVIDIA A100 80GB PCIe, 00000000:25:00.0, 525.85.12, 34, 25 %, 2 %
2023/10/23 21:42:55.237, NVIDIA A100 80GB PCIe, 00000000:81:00.0, 525.85.12, 32, 23 %, 1 %
2023/10/23 21:43:00.238, NVIDIA A100 80GB PCIe, 00000000:25:00.0, 525.85.12, 34, 25 %, 2 %
2023/10/23 21:43:00.240, NVIDIA A100 80GB PCIe, 00000000:81:00.0, 525.85.12, 33, 25 %, 1 %
2023/10/23 21:43:05.241, NVIDIA A100 80GB PCIe, 00000000:25:00.0, 525.85.12, 34, 25 %, 2 %
2023/10/23 21:43:05.243, NVIDIA A100 80GB PCIe, 00000000:81:00.0, 525.85.12, 33, 24 %, 1 %
2023/10/23 21:43:10.244, NVIDIA A100 80GB PCIe, 00000000:25:00.0, 525.85.12, 35, 26 %, 2 %
2023/10/23 21:43:10.246, NVIDIA A100 80GB PCIe, 00000000:81:00.0, 525.85.12, 33, 25 %, 2 %
2023/10/23 21:43:15.247, NVIDIA A100 80GB PCIe, 00000000:25:00.0, 525.85.12, 35, 26 %, 2 %
2023/10/23 21:43:15.248, NVIDIA A100 80GB PCIe, 00000000:81:00.0, 525.85.12, 33, 26 %, 2 %
```

GPU monitoring (AMD gpu)

```
@xsede.org@login-ci1 workshop_amd_gpu]$ squeue --me
  JOBID PARTITION    NAME    USER ST       TIME  NODES NODELIST(REASON)
   5187143   acompile acompile kfotso@x  R    10:03      1 c3cpu-c11-u3-4
   5187117     ami100 sinterac kfotso@x  R    20:16      1 c3gpu-c2-u17
[ @xsede.org@login-ci1 workshop_amd_gpu]$
```



We look for a node
where the GPU job
is running

GPU monitoring (AMD gpu)

```
@xsede.org@c3gpu-c2-u17 ~]$ watch -n 1 rocm-smi --showmeminfo vram
```



This shows that we are
on the gpu node

GPU monitoring (AMD gpu)

```
@xsede.org@c3gpu-c2-u17 ~]$ watch -n 1 rocm-smi --showmeminfo vram
```



This rocm-smi is the
equivalent of nvidia-smi

GPU monitoring (AMD gpu)

```
every 1.0s: rocm-smi --showmeminfo vram
```

```
=====ROCM System Management Interface=====
GPU[1]      : vram Total Memory (B): 34342961152
GPU[1]      : vram Total Used Memory (B): 2590732288
GPU[2]      : vram Total Memory (B): 34342961152
GPU[2]      : vram Total Used Memory (B): 7012352
GPU[3]      : vram Total Memory (B): 34342961152
GPU[3]      : vram Total Used Memory (B): 7012352
=====
=====End of ROCm SMI Log =====
```



We are on GPU 1 and we can tell that the job is using the GPU memory

Questions?